

# *Case Study on the Analysis of SDLC Models Applied to a Real-World Software Application*

**Application:** AI Study Assistant

## **Aim**

To analyze the SDLC models used in developing an AI Study Assistant and identify the most suitable model.

## **Objectives**

- Study SDLC models
- Analyze AI Study Assistant development
- Compare SDLC models
- Recommend the best model

## **Overview**

AI Study Assistant helps students upload notes, ask AI questions, generate quizzes, track progress, and receive study recommendations.

## **Functional Requirements**

| Module    | Features                                              |
|-----------|-------------------------------------------------------|
| Student   | Register, Login, Upload PDFs, AI Chat, Quiz, Progress |
| AI Engine | Summaries, Flashcards, Recommendations                |
| Admin     | Manage users, Reports, Analytics                      |

## SDLC Models Analysis

| Model     | Suitability | Advantages      | Limitations         |
|-----------|-------------|-----------------|---------------------|
| Waterfall | Low         | Simple          | Hard to change      |
| Prototype | High        | User validation | Extra effort        |
| Spiral    | High        | Risk handling   | Costly              |
| Agile     | Very High   | Iterative       | Needs collaboration |
| DevOps    | Very High   | CI/CD           | Infra complexity    |

## Agile Development Process

Requirement Gathering

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Sprint Planning

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System Design

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Development

↓

Testing

↓

Deployment

↓

Student Feedback

↓

Next Sprint

## Requirement Analysis:

Details of Requirement Analysis for the AI Study Assistant.

## System Design:

Details of System Design for the AI Study Assistant.

## Development:

Details of Development for the AI Study Assistant.

## **Testing:**

Details of Testing for the AI Study Assistant.

## **Deployment:**

Details of Deployment for the AI Study Assistant.

## **Maintenance:**

Details of Maintenance for the AI Study Assistant.

## **Comparison of SDLC Models:**

Agile plus DevOps seems to be the best option because of the constant need for upgrades, improvements for the AI, and student feedback.

## **Results:**

The application would benefit from being able to constantly test and update.

## **Conclusion:**

Agile with DevOps seems to be the most ideal methodology when it comes to an AI Study Assistant.